THE PRODUCT OF TWO ALGEBRAIC SPACES IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This candidate was dropped by the mechanical prefilter because of the phrase 'It is clear that H = F x G is a sheaf' at proof line 1119, which is one of three in-place verifications that make up the proof.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Products of representable transformations are representable. Lemma lemma-product-representable-transformations (409-420, produced as L44195), used at proof lines 1119-1122. Makes the diagonal of F x G representable.
- Products of representable transformations with a property. Lemma lemma-product-representable-transformations-property (860-872), used at proof lines 1122-1126. Makes the product chart surjective and etale.

Source reference. Original lemma 02WC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3799–3799. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3809–3809. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F, G be algebraic spaces over S. Then $F \times G$ is an algebraic space, and is a product in the category of algebraic spaces over S.*

Proof. It is clear that $H = F \times G$ is a sheaf. The diagonal of H is simply the product of the diagonals of F and G. Hence it is representable by Lemma 02WC. Finally, if $U \to F$ and $V \to G$ are surjective étale morphisms, with $U, V \in \mathrm{Ob}((Sch/S)_{fppf})$, then $U \times V \to F \times G$ is surjective étale by Lemma 02WM. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.